

B.Tech Capstone Project Proposal

Explainable Chest X-Ray Report Generation with Built-in Hallucination Detection

Degree	Bachelor of Technology, Computer Science and Engineering (AI & ML)
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Abstract

Automated radiology report generation using deep learning can reduce radiologists' workload, but AI-generated reports can describe findings that are not actually present in the image, or that contradict a patient's known clinical history — failure modes broadly known as **hallucination**. This project proposes a chest X-ray report generation system that combines a CNN-based image encoder, a Retrieval-Augmented Generation (RAG) pipeline over prior cases and radiology guidelines, a medical vision-language model (e.g., LLaVA-Med / Med-PaLM 2 / Qwen-VL-Medical) for report generation, and an independently trained multi-label findings classifier. Generated reports are cross-checked in two ways: against the classifier's own predicted findings (image-unsupported hallucinations), and against the patient's documented history — age, sex, prior reports, and lab values (clinical inconsistency hallucinations). Grad-CAM visualizes the image regions responsible for each predicted finding, making the system explainable end-to-end. The system is developed and evaluated on the publicly available IU (Indiana University) Chest X-Ray dataset, supplemented with a small retrieval corpus of radiology reference guidelines.

1. Introduction & Motivation

Writing radiology reports is time-consuming and susceptible to error, particularly for inexperienced radiologists or in resource-constrained hospital settings. Deep learning-based automatic report generation has therefore become an active research area, aiming to support (not replace) radiologists by drafting reports for review. A critical safety concern with AI-generated medical text is hallucination: a model generating fluent, confident-sounding descriptions of findings that are not actually supported by the image, or that ignore what is already known about the patient. This project is motivated by the idea that grounding report generation in retrieved medical knowledge and in the patient's own clinical history — and explicitly checking for both types of hallucination — produces a system that is both more accurate and more trustworthy.

2. Problem Statement

Existing student-level and academic chest X-ray projects typically generate reports from image features alone, using only the current X-ray with no access to medical guidelines, prior similar cases, or the patient's own clinical history. This project addresses the following problem: can a chest X-ray report generation pipeline be made more clinically grounded and self-checking by (i) retrieving relevant prior cases and guidelines before generation, and (ii) validating generated findings against both the image itself and the patient's documented history?

3. Objectives

- To design and train a CNN-based multi-label classifier that predicts clinically relevant findings (e.g., Cardiomegaly, Effusion, Pneumonia, No Finding) from chest X-ray images.
- To build a Retrieval-Augmented Generation (RAG) pipeline that retrieves the top-3 most similar prior cases and relevant radiology guideline excerpts from a vector database before report generation.
- To integrate a medical vision-language model (LLaVA-Med, Med-PaLM 2, or Qwen-VL-Medical) that generates free-text radiology reports conditioned on the image, the retrieved context, and the patient's history.

- To incorporate patient history — age, sex, previous reports, and lab values — as structured text input to the report generation model.
- To design a hallucination detection module with two checks: (i) an image-consistency check comparing the report against the classifier's independent predictions, and (ii) a clinical inconsistency check comparing new findings against the patient's documented history.
- To integrate Grad-CAM-based visual explanations so that every predicted finding is paired with a heatmap showing the supporting image region.
- To evaluate the system using text-generation metrics (BLEU, ROUGE), classification metrics (AUC, F1), and precision/recall for both hallucination-flagging mechanisms.

4. Literature Review

The literature review draws on five areas: CNN-based chest X-ray classification, Transformer-based report generation, medical vision-language models with retrieval augmentation, visual explainability, and hallucination detection in medical AI. Full citations with links are provided in Section 12 (References).

Classification backbone: Rajpurkar et al. (CheXNet) and Irvin et al. (CheXpert) – established CNN-based (DenseNet121) multi-label chest X-ray classification, which this project's classifier branch builds upon.

Report generation architecture: Vaswani et al. (Transformer) and Chen et al. (R2Gen) – the base CNN-encoder plus Transformer-decoder design this project extends.

Medical vision-language models & retrieval augmentation: Li et al. (LLaVA-Med) and Singhal et al. (Med-PaLM 2) – large medically-aligned vision-language models capable of clinical reasoning; Lewis et al. (RAG) – the retrieval-augmented generation framework used to ground the medical LLM's output in retrieved cases and guidelines rather than relying on parametric memory alone.

Explainability: Selvaraju et al. (Grad-CAM) – the gradient-based visual explanation technique used to localize predicted findings on the X-ray.

Hallucination detection: Li et al. and Chen et al. (Med-HallMark), along with the MedHEval benchmark – recent work studying and detecting hallucinations in medical vision-language models, forming the basis for this project's dual hallucination-checking design.

5. Identified Gaps in Existing Work & Proposed Enhancements

Based on faculty review of the baseline design, two specific gaps were identified in typical CNN-plus-Transformer report generation pipelines, along with corresponding enhancements adopted into this project.

Gap 1

Plain Transformer decoders generate reports purely from image features. They have no access to established medical guidelines or prior similar cases, so their output is not grounded in external clinical knowledge and can be generic or ungrounded.

Enhancement

Introduce a medical vision-language model — LLaVA-Med, Med-PaLM 2, or Qwen-VL-Medical — combined with Retrieval-Augmented Generation (RAG). Before generating a report, the system

retrieves the top-3 most similar prior cases and relevant radiology guideline excerpts from a vector database. This retrieved context, along with the image, is passed to the medical LLM for report generation. The findings classifier is then used, as before, to cross-check the generated report and flag hallucinations.

Gap 2

The baseline pipeline uses only the current X-ray image, with no awareness of the patient's broader clinical context, so it cannot detect findings that contradict what is already known about the patient.

Enhancement

Feed the patient's history — age, sex, previous reports, and lab values — into the report generation model as additional text input. The hallucination detection module then checks whether newly reported findings contradict the patient's history and, if so, flags them as a **clinical inconsistency hallucination** — a category distinct from image-unsupported hallucinations. For example, if the generated report states “new fracture” while the patient's history records that a cast was removed two years earlier, this contradiction is flagged rather than passed through silently.

6. Proposed Methodology / System Architecture

6.1 Overall Pipeline

- **Inputs:** the chest X-ray image, and the patient's history (age, sex, previous reports, lab values) as structured text.
- **Image Encoder (DenseNet121):** extracts visual features from the X-ray.
- **Retrieval Module:** queries a vector database of prior cases and radiology guideline excerpts, returning the top-3 most relevant items given the current image/history.
- **Medical LLM Decoder (RAG-augmented):** a medical vision-language model (LLaVA-Med / Med-PaLM 2 / Qwen-VL-Medical) generates the free-text report from the image features, the retrieved context, and the patient history.
- **Findings Classifier:** an independent multi-label head predicts structured findings directly from the image.
- **Hallucination Detection (two checks):** (i) report findings vs. classifier predictions → image-unsupported hallucination; (ii) report findings vs. patient history → clinical inconsistency hallucination.
- **Grad-CAM:** visualizes the image region supporting each classifier-predicted finding.
- **Output:** generated report, predicted findings, both categories of flagged hallucinations, and the Grad-CAM overlay.

6.2 Module Summary

Module	Function	Core Technique
Image Encoder	Extract visual features from the X-ray	DenseNet121 (pretrained, fine-tuned)
Retrieval Module	Retrieve top-3 similar cases & guideline excerpts	Vector database (FAISS / ChromaDB) + embeddings
Medical LLM Decoder	Generate free-text report, grounded in retrieved context and patient history	LLaVA-Med / Med-PaLM 2 / Qwen-VL-Medical (RAG-augmented)
Findings Classifier	Predict structured clinical findings	Multi-label classification head
Hallucination Checker	Flag image-unsupported and clinically inconsistent findings	Classifier cross-check + patient-history contradiction check
Explainability	Localize predicted findings on the image	Grad-CAM

7. Dataset

IU (Indiana University) Chest X-Ray Dataset (distributed via the NLM Open-i public repository) — approximately 7,400 frontal/lateral chest X-ray images paired with their original radiology reports and associated finding labels, used to train and evaluate the image encoder, classifier, and report generation model. This is supplemented with two additional components required for the enhanced design: (i) a small retrieval corpus of publicly available radiology reference guidelines, used to populate the vector database; and (ii) patient history fields, derived from the comparison/prior-study information already

present in the IU X-Ray reports where available, since structured EHR data is not part of any public chest X-ray dataset.

8. Tools & Technologies

Category	Tools / Libraries
Programming Language	Python
Deep Learning Framework	PyTorch / TensorFlow
CNN Backbone	DenseNet121 (torchvision / Keras Applications)
Medical Vision-Language Model	LLaVA-Med / Med-PaLM 2 / Qwen-VL-Medical (frozen inference or lightweight LoRA fine-tuning)
Vector Database & Embeddings	FAISS / ChromaDB with Sentence-Transformers or BioBERT embeddings
Explainability	Grad-CAM (pytorch-grad-cam or custom implementation)
Evaluation	NLTK / pycocoevalcap (BLEU, ROUGE), scikit-learn (AUC, F1)
Demo Interface	Streamlit or Flask
Compute Environment	Google Colab / Kaggle Notebooks (free-tier GPU)

9. Proposed Timeline

Phase	Weeks	Deliverable
Literature review & dataset preparation	1-2	Finalized literature review, cleaned dataset
Findings classifier development	3-5	Trained multi-label classifier + Grad-CAM module
Vector database & RAG pipeline setup	6-7	Guideline/case corpus indexed; top-3 retrieval working
Medical LLM integration (report generation)	8-11	Reports generated using image + retrieved context + patient history
Hallucination detection module (both checks)	12-13	Working image-consistency and clinical-inconsistency checkers
Integration & demo interface	14-15	End-to-end demo (image + history to report to flags)
Evaluation, documentation & report writing	16-18	Final evaluation results and project report

10. Expected Outcomes

- A trained multi-label chest X-ray findings classifier with reported per-class AUC/F1 scores.
- A working RAG pipeline retrieving relevant prior cases and guideline excerpts to ground report generation.
- A medical-LLM-based report generator producing free-text radiology reports, evaluated using BLEU/ROUGE.
- A dual hallucination-detection mechanism with measured precision/recall for both image-unsupported and clinical-inconsistency flags.
- Grad-CAM visualizations demonstrating explainable predictions for each finding.
- A functional demo application illustrating the complete pipeline end-to-end.

11. Ethical Considerations

This project is a research and educational prototype and is **not intended for clinical or diagnostic use**. Only fully de-identified, publicly released academic data is used; no real patient records or EHR data are collected or processed. This disclaimer will be clearly stated in the final report and any demonstration of the system.

12. References

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Note: References are limited to peer-reviewed conference proceedings and established preprints (IEEE/NeurIPS/ACL/AAAI venues); journal-only sources have been excluded.